PROJECT QUICK-START & VIVA GUIDE

MediCare Hospital Management System

Complete Documentation for Running, Explaining, and College Viva Submission

1. Login System & Role-Based Access (Kaun Login Karega?)

System mein **Role-Based Access Control (RBAC)** set hai taaki galat access na mile. Database mein default 2 accounts bane hue hain:

Role	Email ID	Password	Responsibilities / Kaam
Administrator	`admin@medicare.com`	`admin123`	Hospital Head / Director: Doctor directory manage karna, patient records check/delete karna, system dashboard dekhna.
Receptionist / Staff	`staff@medicare.com`	`staff123`	Frontdesk Counter: OPD aane wale naye patient register karna aur unke liye doctor slot par appointment book karna.

*Note: Passwords direct text mein nahi, **Bcrypt hashing** se encrypt hain aur authentication **JWT (JSON Web Token)** ke through secure hai.*

2. Project Kaise Kaam Karta Hai? (Step-by-Step Flow)

> **Step 1: Patient OPD Entry**
> Naya patient hospital aata hai → Receptionist **Patients Tab** mein jata hai → Name, Age, Gender, Phone, Blood Group daal kar register karta hai.

> **Step 2: Specialist Doctor Selection**
> Hospital ke paas **Doctors Tab** mein doctors ki list hoti hai (Cardiologist, Pediatrician, Orthopedic, consulting fees aur schedule).

> **Step 3: Appointment Booking**
> **Appointments Tab** mein jakar registered patient aur doctor select karke desired date aur time-slot par booking generate ki jaati hai.

> **Step 4: Real-time Status Update**
> Consultation ke baad appointment ka status `Scheduled` se badal kar `Completed` ya `Cancelled` kiya ja sakta hai.

> **Step 5: Live Dashboard Overview**
> **Dashboard Tab** par Total Patients, Active Doctors, Scheduled Appointments aur Pending Visits ke counters real-time update hote hain.

3. Dost ke PC Par Kaise Run Karein? (Setup Guide)

Yeh project bina kisi MySQL ya XAMPP server ke chalta hai kyunki isme embedded **SQLite Database** hai jo project folder ke andar hi rehta hai.

1st Time Setup (Sirf ek baar karna hai):

1. Dost ke PC mein **Node.js** install hona chahiye.
2. `backend` folder mein terminal khol kar run karo: `npm install`
3. `frontend` folder mein terminal khol kar run karo: `npm install`

Everyday Launch:

Bas root folder mein `run-project.bat` par **Double-Click** kar do! Dono servers (Backend 5000 + Frontend 5173) start ho jayenge aur browser direct khul jayega.

4. College / IGNOU Viva Me Kaise Explain Karein?

Examiner Viva Quick-Script (Short & Confident):

- **Problem & Solution:** "Traditional OPD register-based records mein record misplace hote hain aur slot clash hota hai. Humne central automated system banaya hai jo real-time triage handle karta hai."
- **Architecture:** "3-Tier Architecture use hui hai: Frontend mein React.js (Vite) with responsive modern UI and smooth scrolling, Backend mein Node.js & Express.js REST APIs, aur Database mein Relational SQLite3 engine (Foreign Key relationships ke saath)."
- **Security & Portability:** "Authentication stateless JWT tokens aur Bcrypt password encryption ke through secured hai. System zero-configuration hai, kisi heavy external database server ki zaroorat nahi padti."